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Unit – 5 : Session – 2 : SLO – 2 | Memory Consistency and Communication Overhead

■ I. Multiple Choice

1. b) A contract that defines the order in which memory operations appear to execute.
2. b) Time spent transferring data between processors or nodes.
3. b) Memory coherence problem — inconsistent views of shared data across processors.
4. c) Parallel execution may become slower than sequential execution — overhead exceeds the benefit.
5. b) Sequential consistency — all processors observe all writes in exactly the same order.

■ II. True or False

1. TRUE — consistency models define a common observed order of memory operations.
2. FALSE — weak models deliberately relax immediate visibility for performance.
3. TRUE — frequent data exchange across the network raises communication overhead.
4. TRUE — lower communication overhead means adding processors keeps paying off (better scalability).
5. FALSE — coherence is about keeping cached copies identical; consistency is about the ordering of operations — related but distinct.

■ III. Scenario-Based Activity

- Issue 1 — inconsistent results across runs: a memory consistency/coherence problem. Processors exchange intermediate results but observe writes in different orders (relaxed visibility), so computations consume stale values; non-deterministic ordering makes each run differ.
- Issue 2 — execution time grows with more processors: communication overhead. With 64 nodes exchanging results frequently, message-passing time grows faster than the per-node computation shrinks, so total time increases — the classic scalability limit (overhead > computation).
- Fixes: enforce synchronization/consistency at exchange points (barriers, locks, release consistency), reduce communication frequency (larger messages, better partitioning), and overlap communication with computation.